Discrete Mathematics

1001

2026 年 10 月 20 日

目录

第一章 Logic and Proofs **5**

1.1 Basic 5

1.2 Logical Operators 6

1.2.1 Defines 6

1.2.2 Rules 6

1.2.3 Normal Forms 7

1.3 Predicates and Quantifiers 7

1.3.1 Defines 7

1.3.2 Rules 7

1.3.3 Prenex Normal Form 8

1.4 Rules of Inference 8

1.4.1 Rules 8

1.4.2 Fallacies 9

1.5 Proof Methods 9

第二章 Relations **11**

2.1 Definitions 11

2.1.1 Representation 11

2.1.2 Special Relations 11

2.1.3 Properties 12

2.1.4 Operations 12

2.2 Closures of Relations 13

2.2.1 Definations 13

2.2.2 Connectivity Relation 13

2.3 Equivalence Relations 13
2.4 Partial Orderings 13

第一章　Logic and Proofs

The Only Truth is the Truth Table.

1.1　Basic

Proposition: A declarative sentence that is either True or False, but not both.
not declarative: $x + 1 = 3$
not T or F: Paradox. e.g. This sentence is false.
allow its truth value be undetermined: He will go to Paris.

Proposition Varible: p, q, r, etc.

Atomic Proposition: the simplest proposition.

Given proposition $p \to q$:

Converse 逆命题	$q \to p$
Inverse 否命题	$\neg p \to \neg q$
Contrapositive 逆否命题	$\neg q \to \neg p$

p: **Hypothesis** / **Antecedent** / **Premise**
q: **Conclusion** / **Consequence**

Tautology: always True;
Contradiction: always False;
Contingency: sometimes True, sometimes False;
Satisfiable: 存在赋值方式使为真, Tautology + Contingency

Precedence: $\forall/\exists > \neg > \wedge/\vee > \to / \leftrightarrow$

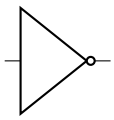
Theorem: a somewhat important statement.
Proposition / Facts / Results: less important.
Axiom 公理 / **Postulate** 假设: can be used as premises.
Lemma: a less important theorem that is helpful for other results.
Corollary 推论: a theorem that can be established directly from a theorem.
Conjecture 猜想: guess to be true.

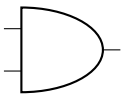
1.2 Logical Operators

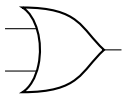
1.2.1 Defines

Negation 否定	$\neg x$ / $!x$ / $\sim x$	not		$1-x$
Conjunction 合取	$x \wedge y$	and		xy
Disjunction 析取	$x \vee y$	or		$x+y-xy$
Exclusive Or	$x \oplus y$	xor		
Conditional / Implication 蕴涵	$x \to y$	if, then	$\neg x \vee y$	$1-x+xy$
Biconditional / Equivalence 等价	$x \leftrightarrow y$	iff	$\neg(x \oplus y)$	

x only if y: $x \to y$
x unless y: $y \to \neg x$
x, y both occur: $x \wedge y$, no causality.
avoid using *any*, 因为自然语言下同时有任意一个/某一个的含义。


Inverter


And Gate


Or Gate

1.2.2 Rules

Logical Equivalence:

- Identity: $x \wedge \mathbf{T} \equiv x$, $x \vee \mathbf{F} \equiv x$
- Domination: $x \wedge \mathbf{F} \equiv \mathbf{F}$, $x \vee \mathbf{T} \equiv \mathbf{T}$
- Idempotent: $x \wedge / \vee x \equiv x$
- Double Negation: $\neg(\neg x) \equiv x$
- Communtative & Associative Laws: $\wedge/\vee$ 分别满足交换 & 结合律
- Distribution: $\wedge/\vee$ 分别对 $\vee/\wedge$ 分配
  $$\begin{cases} x \vee (y \wedge z) & \equiv (x \vee y) \wedge (x \vee z) \\ x \wedge (y \vee z) & \equiv (x \wedge y) \vee (x \wedge z) \end{cases}$$
- De Morgan's Laws: flip operator
  $$\begin{cases} \neg(x \wedge y) & \equiv \neg x \vee \neg y \\ \neg(x \vee y) & \equiv \neg x \wedge \neg y \end{cases}$$
- Absorption Laws:
  $$\begin{cases} x \vee (x \wedge y) & \equiv x \\ x \wedge (x \vee y) & \equiv x \end{cases}$$

1.2.3 Normal Forms

Simple Conjunctive/Disjunctive Formula: only consists of $\wedge/\vee$ and $\neg$

Conjunctive/Disjunctive Normal Form (CNF / DNF): $\bigwedge(\vee)$ / $\bigvee(\wedge)$
主合取/析取范式：外层为合取/析取。CNF / DNF 必存在，构造性证明，略。

CNF / DNF 不唯一，故引出 **Principle/Major C/D NF** (P*NF/M*NF)（主合/析取范式），强制约定每个基本项包含所有 n 个变元。 P*NF 是唯一的。

极小项：$\bigwedge_{i=1}^{n}$, $m_0 \sim m_{2^n-1}$；
极大项：$\bigvee_{i=1}^{n}$, $M_0 \sim M_{2^n-1}$；
编号：from left to right, i-th bit $= [p_i]$。
e.g. $(p_1 \wedge \neg p_2 \wedge \neg p_3) = m_{100} = m4$; $(\neg p_1 \vee p_2) = M_{01} = M_1$

Construct P*NF base on *NF: for each simple formula with less than n varibles, adjust, and merge the same formulas
a. $A \equiv (A \wedge p_k) \vee (A \wedge \neg p_k)$
b. $A \equiv (A \vee p_k) \wedge (A \vee \neg p_k)$

转换 PCNF/PDNF，其中 $\sim$ 为按位取反：$\bigwedge\limits_{t\in S} M_t = \bigvee\limits_{t\in U\setminus S} m_{\sim t}$； $\bigvee\limits_{t\in S} m_t = \bigwedge\limits_{t\in U\setminus S} M_{\sim t}$
证明：PDNF $\to$ PCNF：每个 $t \in S$ 表示一种成真赋值，则每个 $t \in U\setminus S$ 表示一种成假赋值。为使结果为真，则每种成假赋值均需有一项不成立，即补集的按位取反。易知该操作可逆，则 PCNF $\to$ PDNF 亦得。

1.3 Predicates and Quantifiers

1.3.1 Defines

Predicate 谓词: refer to a property, propositonal function P, $P(x_1, \cdots x_n)$

Quantifier 量词: the extent to which P is true over a range of elements
a. universal quantifier 全称量词: $\forall$
b. existential quantifier 存在量词: $\exists$
c. uniqueness quantifier 唯一量词: $\exists!/\exists_1$

Domain: the domain of x; the domain consists of $a, b, c, \cdots$
Scope: the scope of $\forall/\exists$, etc.

Bound Occurance 约束出现: x is restricted by quantifiers.
Free Occurance 自由出现: x is not restricted by any quantifiers.
Closed Form 闭式: no free varibles, all varibles is in some scope.

1.3.2 Rules

- Negation:
  a. $\neg\forall \equiv \exists\neg$
  b. $\neg\exists \equiv \forall\neg$

- Distribution: turn $\rightarrow$ / $\leftrightarrow$ to $\wedge/\vee$ first
  a. under the same restriction: only the following two are correct
  - $\forall x(A(x) \wedge B(x)) \equiv \forall x A(x) \wedge \forall x B(x)$
  - $\exists x(A(x) \vee B(x)) \equiv \exists x A(x) \vee \exists x B(x)$

  b. under different restriction: $\forall/\exists x(A(x) \wedge / \vee B) \equiv \forall/\exists x A(x) \wedge / \vee B$

1.3.3 Prenex Normal Form

前束范式: all quantifier are at the beginning, and their scopes last to the end.
e.g. $\forall x \exists y \forall z(P(x,y) \vee Q(x,y,z) \vee \neg R(z))$

PNF 必存在，构造性证明，略。
PNF 不唯一。

1.4 Rules of Inference

a **valid argument**: ($\bigwedge$ premises) $\rightarrow$ conclusion

fallacy: 谬误

1.4.1 Rules

Modus Ponens	肯定前件	$\begin{cases} x \\ x \rightarrow y \end{cases} \implies y$
Modus Tollens	否定后件	$\begin{cases} \neg y \\ x \rightarrow y \end{cases} \implies \neg x$
Hypothetical Syllogism	假言三段论	$\begin{cases} x \rightarrow y \\ y \rightarrow z \end{cases} \implies x \rightarrow z$
Disjunctive Syllogism	析取三段论	$\begin{cases} x \vee y \\ \neg x \end{cases} \implies y$
Addition		$x \implies (x \vee y)$
Simplification		$(x \wedge y) \implies x$
Conjunction		$\begin{cases} x \\ y \end{cases} \implies (x \wedge y)$
Resolution	消解	$\begin{cases} (x \vee y) \\ (\neg x \vee z) \end{cases} \implies (y \vee z)$

Constructive Dilemma		$\begin{cases}(x \to y) \land (u \to v) \\ x \lor u\end{cases} \implies (y \lor v)$
Destructive Dilemma		$\begin{cases}(x \to y) \land (u \to v) \\ (\neg y \lor \neg v)\end{cases} \implies (\neg x \lor \neg u)$
Universal Instantiation (UI)		$\forall x P(x) \implies P(u)$
Existential Instantiation (EI)		$\exists x P(x) \implies P(u)$ for some u
Universal Generalization (UG)		$P(u)$ for an arbitrary $u \implies \forall x P(x)$
Existential Generalization (EG)		$P(u)$ for some $u \implies \exists x P(x)$
Universal Modus Ponens		$\begin{cases}\forall x(P(x) \to Q(x)) \\ P(u)\end{cases} \implies Q(u)$
Universal Modus Tollens		$\begin{cases}\forall x(P(x) \to Q(x)) \\ \neg Q(u)\end{cases} \implies \neg P(u)$

prep.: using / from

e.g.

1.	$\neg t$	premise 3.
2.	$r \to t$	premise 2.
3.	$\neg r$	modus tollens using 1, 2
4.	$\neg r \lor \neg s$	addition from 3
5.	$\neg(r \lor s)$	De Morgan's

1.4.2 Fallacies

- fallacy of affirming the conclusion:
  $\begin{cases}x \to y \\ y\end{cases} \not\Rightarrow x$
- fallacy of denying the hypothesis:
  $\begin{cases}x \to y \\ \neg x\end{cases} \not\Rightarrow \neg y$

1.5 Proof Methods

Direct Proof.

Conditional Proof (CP) 附加前提证明: to prove $A \implies (B \to C)$, use B as a premise. try to prove $(A \land B) \to C$.

Proof by Contraposition: use $\neg q \to \neg p$ to prove $p \to q$.

Proof by Contradiction 反证

Proof by Counterexample 找反例: use $\neg P(u)$ to prove $\neg\forall x P(x)$.

Existence Proof:
a. Constructive: find a witness, that $P(u)$ is true.
b. Nonconstructive

Exhaustive Proof / Proof by Exhaustion 穷举
Proof by Cases 分讨

Tips:

- Vacuous Proof: if p is false, then $p \to q$ is immediately true.
- Trivial Proof: if q is true, then $p \to q$ is always true.
- proof of equivalence:
  a. to prove $p \leftrightarrow q$, proving $(p \to q) \wedge (q \to p)$.
  b. to prove $p_1 \leftrightarrow p_2 \leftrightarrow \cdots \leftrightarrow p_n$, proving $\bigwedge(p_i \to p_{i \bmod n+1})$.
- **WLOG**: without loss of generality.

第二章　Relations

Can't use emptyset for Cartesian Product.

2.1 Definitions

A binary relation R from A to B is a subset of $A \times B$: $R \subset A \times B$

If $(a, b) \in R$, then a is related to b. The **order** matters. Denoted as aRb or $(a, b) \in R$. Otherwise, $a \not R b$ or $(a, b) \notin R$.

The length of tuple is called degree. e.g. The degree of $A \times \cdots \times A$ is n.
A_i is called the domains of R.

2.1.1 Representation

Order!

- **directed** bipartite graph
- **directed** graph for $R \subset A \times A$
- 01-matrix: $R : A \to B$, $M_{i,j} = [(a_i, b_j) \in R]$
  $M_{R \cap S} = M_R \wedge M_S$; $M_{R \cup S} = M_R \vee M_S$
  $M_{S \circ R} = M_R \odot M_S$ a.k.a. $M_{i,j} = \vee(M_{S_{i,k}} \wedge M_{R_{k,j}})$ **矩阵顺序不一样!**
  $M_{R^n} = M_R^n$

2.1.2 Special Relations

- 空关系 $\emptyset$
- 恒等关系 $I_A = \{(a, a) : \forall a\}$
- 全域关系 $U_A = \{(a, b) : \forall a \forall b\}$

2.1.3 Properties

- 自反性 Reflexive: $\forall x \in A, xRx$
  Each node has a self-loop.
  All elements on the main diagonal are 1.

- 反自反性 Irreflexive: $\forall x \in A, (x, x) \notin R$
  Irreflexive is **not** the negation of reflexive !
  R can be neither reflexive nor irreflexive. R is both reflexive and irreflexive iff $A = \emptyset$ (however, Cartesian Product asks $A \neq \emptyset$ in $A \times A$, so?).
  Each node must not have a self-loop.
  All elements on the main diagonal are 0.

- 对称性 Symmetric: $R = R^{-1}$ (a.k.a. $\forall x, y \in A\ (x, y) \in R \iff (y, x) \in R$)
  （等价地）All edges are bi-directed.
  All are same across the main diagonal.

- 反对称性 Antisymmetric: $(x, y) \in R \wedge (y, x) \in R \implies x = y$
  Antisymmetric is **not** "not symmetric" !
  For any two different vertices, there are at most one directed edge link them.
  All 1's are across from 0's.

- 非对称性 Asymmetric: $(x, y) \in R \implies (y, x) \notin R$
  Completely no symmetric. Asymmetric $\iff$ Antisymmetric $\wedge$ Irreflexive.
  No self-loops, and for any two different vertices, there are at most one directed edge link them.
  All elements on the main diagonal are 0, and all 1's are across from 0's.

- 传递性 Transitive: $(x, y), (y, z) \in R \implies (x, z) \in R$

2.1.4 Operations

Inverse Relation: $R^{-1} := \{(b, a) : (a, b) \in R\}$.
e.g. $R = \{(a, b) : a < b\}$, then $R^{-1} = \{(a, b) : a > b\}$.
Note that if $R = \{(a, b) : p(a, b)\}$, R^{-1} is **not** $\{(a, b) : \neg p(a, b)\}$!

Properties of Inverse Relation:
a. $(R^{-1})^{-1} = R$
b. $(R\ \cap / \cup / -\ S)^{-1} = R^{-1}\ \cap / \cup / -\ S^{-1}$
c. $R \subset S \iff R^{-1} \subset S^{-1}$
d. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$

Composition: $S \circ R : A \overset{R}{\to} B \overset{S}{\to} C$

Power:
a. $R^0 := I$
b. $R^n := R^{n-1} \circ R$
c. $R^{-n} := (R^{-1})^n$

Thm. R is transitive iff $R^n \subset R$ for all $n \in \mathbb{N}_+$

Thm. Composition of relations satisfies the associative law.

2.2 Closures of Relations

2.2.1 Defination

For any property P, the "P closure" of R, if it exists, is defined as the "smallest" subset of $A \times A$ containing R with P.

- contains R
- has property P
- "smallest": for every $R \subset T$ s.t. T has property P, $S \subset T$.
  因此，闭包是合法集合，且是所有合法 T 的交集。若把集合的包含关系看作合法集之间的偏序关系，则存在闭包说明存在**唯一**的极小元，这对 P 本身有所限制。

closures:

- reflexive closure: $r(R) = R \cup I_A$
- symmetric closure: $s(R) = R \cup R^{-1}$
- transitive closure: $t(R) = R^* = \bigcup\limits_{n \in \mathbb{N}_+} R^n$

2.2.2 Connectivity Relation

a.k.a. transitive closure

$(a, b) \in R^n$: there exists a path from a to b with length n

$|A| = n$, if there is a path $a \rightsquigarrow b$, then there must exist a path with length $\leqslant n$. Therefore, $t(R) = R \cup \cdots \cup R^n$

- naïve algo: brute force $O(n^4)$
- Warshall's: $w_{i,j} \leftarrow w_{i,k} \wedge w_{k,j}$, $O(n^3)$
  interior vertices: exclude a, b

2.3 Equivalence Relations

2.4 Partial Orderings